Foundations of Statistical Learning & Algorithms

Today's goal • 3/3 items

Module 1
Introduction to Linear Algebra and Vector Spaces

Module 2
Linear Transformation and Matrix Decomposition

Linear Independence
- Understanding Linear Independence in Vector Spaces — Reading • 18 min
- Linear Independence, Basis, and Rank — Ungraded App Item • 20 min
- Linear Independence — Reading • 30 min

Linear Mapping
- Exploring Transformations: Understanding Linear Mappings — Reading • 1h
- Linear Mapping: Part 1 — Video • 8 min
- Linear Mapping: Part 2 — Video • 8 min
- Linear Mapping — Reading • 5 min
- Matrix Representation — Reading • 6 min
- Practice Quiz: Linear Mapping — Graded Assignment • Grade: 100%

Lesson 3: Eigenvalues and Eigenvectors
- Introduction to Eigenvalues and Eigenvectors — Reading • 40 min
- Eigenvectors and Eigenvalues: Examples and Applications — Reading • 30 min

Lesson 4: Cholesky Decomposition
- **Cholesky Decomposition** — Reading • 5 min
- Solved Examples — Reading • 10 min

Lesson 5: Singular Value Decomposition
- Introduction to Singular Value Decomposition (SVD) — Reading • 28 min
- Derivation of Singular Value Decomposition (SVD) from Eigenvalues and Eigenvectors — Reading • 15 min

Module 3
Probability Foundations for Statistical Learning

Module 4
Introduction to Model Evaluation and Optimization

# Cholesky Decomposition

1:15 / 1:25 2x

T**he Cholesky Factorization Theorem:** States that any symmetric, positive definite matrix A can be decomposed into the product A=LL^T, where L is a lower triangular matrix with positive diagonal elements.

$$\begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{nn} \end{bmatrix} = \begin{bmatrix} l_{11} & \cdots & 0 \\ \vdots & \ddots & \vdots \\ l_{n1} & \cdots & l_{nn} \end{bmatrix} \begin{bmatrix} l_{11} & \cdots & l_{n1} \\ \vdots & \ddots & \vdots \\ 0 & \cdots & l_{nn} \end{bmatrix}$$

$L$ is called the Cholesky factor of $A$, and $L$ is unique.

**Applications:**

- **Covariance Matrices in Gaussian Distributions:** The Cholesky decomposition is used to transform a standard normal variable into a correlated normal variable.
- **Efficient Computation of Determinants:** For a matrix A, det(A)=det(L)^2, where det(L) is the product of the diagonal elements of L.
- **Solving Systems of linear equations:** If A is Symmetric Positive Definite (SPD), the Cholesky factorization is employed to solve the linear system Ax = y. When the components are substituted into the equation, the result is $LL^Tx = y$. Assuming $z = L^Tx$,Thus, the desired answer z can be estimated by resolving the triangular system of equations Lz = y, and x may be computed by calculating the triangular linear system $L^Tx = z$.

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